

Daniel Yao-Ting Huang

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Education

University of California, San Diego

La Jolla, CA

M.S. in Electrical and Computer Engineering(Intelligent Systems and Robotics/Control)

Sep. 2024 - Jun. 2026(Expected)

- Overall GPA: 3.55/4.0
- Selected Courses: Sensing & Estimation Robotics, Planning & Learning Robotics, Robot Manipulation and Control

National Taiwan University(NTU)

Taipei, Taiwan

B.S. in Electrical Engineering

Sep. 2019 - Aug. 2023

- Overall GPA: 3.71/4.3(3.64/4.0) Last 60 GPA: 3.82/4.3(3.75/4.0)
- CS-related GPA: 4.0/4.3(**3.94/4.0**)
- **Teaching Assistant** for Cornerstone EECS Design and Implementation (2023 Spring) - Instructor: Prof. Ho-Lin Chen
- Selected Courses: Robotics*, Machine Learning, Deep Learning for Computer Vision*. (*indicates graduate courses)

Publications

The N2D Haptic Glove: A Multi-Finger Glove for 2D Directional Force Feedback for Contact Rich Manipulation

Yao-Ting Huang, Kaitlin Calimbahin, Jake Honma, Logan Li, Omar Hernandezand, Michael Yip

(2025). 2025, under review in International Conference on Intelligent Robots and Systems(IROS'26)

In-Hand Manipulation of Articulated Tools with Dexterous Robot Hands with Sim-to-Real Transfer

Soofiyan Atar, **Daniel Huang**, Florian Richter, Michael Yip

arXiv preprint arXiv:2509.23075 (2025). 2025, under review in International Conference on Intelligent Robots and Systems(IROS'26)

Research Experience

Advanced Robotics and Controls Lab (ARCLab), UC San Diego

Michael C. Yip

Graduate research student

Oct 2024 – Present

- Built a **visuotactile teleoperation and data pipeline** connecting a haptic glove, Inspire Hand, and Franka arm, with synchronized multimodal sensing and control across simulation and hardware.
- Trained **tactile-conditioned imitation learning policies** using **LeRobot ACT** and **Diffusion Policy**, improving sample efficiency and task success over vision-only baselines.
- Implemented **closed-loop 2-D force-feedback teleoperation**, improving force-tracking accuracy and reducing operator workload in contact-rich manipulation.
- Built a **ROS 2-based motor torque-sensor calibration system** for Inspire Hand and Wuji Hand using an **ATI load cell** for ground-truth measurement and **Rerun** for visualization and debugging.

Robot Learning Lab, NTU

Shau-Hua, Sun

Undergraduate research student

July 2022 – Dec 2023

- Investigated Unsupervised Reinforcement Learning (RL) problems based on SPIRL.
- Designed experiments showing that the learning efficiency will be slower by **1M** steps due to the absence of certain skills.
- Implemented choice-mechanism that allows high-level agent to choose between exploration and exploitation.

Advanced Control Lab, NTU

Li-Chen, Fu

Undergraduate research student

Feb 2022 – June 2022

- Assisted the experiment of the UAV (unmanned aerial vehicle) system in the indoor environment and anchor setup.
- Conducted experiments and developed a **2-D visual odometry** system to achieve indoor UAV odometry in GPS-denied environments using **optical flow** techniques.

Work Experience

Industrial Technology Research Institute(Intern)

Hsinchu, Taiwan

Reinforcement Learning, Autonomous system, ROS, Docker

Aug. 2023 - Dec. 2023

- Contributed to self-driving automobile group by applying reinforcement learning for behavioral prediction of autonomous vehicles.
- Designed a virtual training environment for reinforcement learning framework based on real-world collected vehicle data.

CAVEDU(Summer Intern)

Taipei, Taiwan

Python, Google AIY, Program Education

Jan. 2016 - Jun. 2017

- Delivered Python/ML workshops; co-authored teaching kits used by **100+** K-12 students.

Selected Projects

Visual-Inertial SLAM

ECE 276A Sensing & Estimation for Robotics

Jan 2025 - Mar 2025

- Built stereo pipeline (Shi-Tomasi/optical-flow/RANSAC) with EKF fusion; update time **<50 ms** on recorded sequences.
- Tuned noise models and outlier rejection; achieved drift decreasing **~80%** vs IMU-only across **3** evaluation runs.
- Implemented robust keyframe logic to sustain tracking under fast motion.

Motion Planning Algorithms

ECE 276B Planning & Learning for Robotics

Apr 2025 - May 2025

- Developed **Weighted A*** and KD-accelerated **Connect-RRT** planners for 3-D obstacle maps; achieved **26×** NN-query speed-up with cKDTTree.
- Evaluated seven maps: Weighted A* (w = 2) ran 23 % faster than optimal A* with only 3 % longer paths.
- Produced comparative study to analyze runtime, memory, and path quality across search-based and sampling-based methods.

Magic Hand — Real-Time RGB-D Arm Pose Guided Laser Pointer.

2023 MakeNTU [Demo]

Apr 2023 - May 2023

- Applied **YOLOv5** on RGB-D frames from **Intel RealSense** camera to localize arm joints and estimate depth for laser-pointer steering in real time.
- Optimized inference with **optical-flow** tracking to reduce per-frame compute and sustain **60+ FPS** on a laptop CPU.
- Delivered an **award-winning demo** with robust, low-latency pointing and smooth user interaction.

Awards & Competitions

2023 **3rd prize of Enterprise prize & Best Application Award**, MakeNTU

Taipei, Taiwan

2021 **3rd prize of Interactive Technology**, NIICC

Taipei, Taiwan

2021 **4th prize of Enterprise prize**, MakeNTU

Taipei, Taiwan

2020 **Special prize of Enterprise prize**, Meichu Hackathon

Shinchu, Taiwan

2020 **Special prize of Enterprise award**, MakeNTU

Taipei, Taiwan

Leadership & Volunteer experience

2022 MakeNTU

Event General Coordinator

Aug 2021 - May 2022

- Orchestrated a national competition with over **34 teams** and **134 undergraduates**, leading marketing and coordination efforts.
- Enhanced event visibility, contributing to a **20%** increase in participant engagement.

Academic Department of NTUEE Student Association

Manager & Course lecturer

Sep 2020 - Aug 2023

- Facilitated the usage and maintenance of machine tools for student projects.
- Managed inventory, ensuring the availability of modules and microcontrollers for over 200 students.
- Delivered lectures on the fundamental use and advanced techniques of **AutoCAD** and **Fusion360**.

Computer program consulting service

Volunteer

Sep 2020 - Jan 2021

- Provided weekly consultation services to assist students from various departments in overcoming challenges with programming languages such as C++ and Python.

Skills

Programming Python, C/C++, C#, MATLAB, JavaScript, Pytorch

Simulation ROS 2, Isaac Lab, Isaac Sim, Mujoco, Unity, Blender

Hardware Jetson Nano, Raspberry Pi, STM32, Arduino, Teensy, Mechanical design(CAD, Onshape)

Robotic Systems Franka Panda Arm, Unitree G1, Inspire Hand, Wuji hand, UAVs